

PHY175: Physics for Engineers & Electronics

100-Percentile Master Syllabus Blueprint, Curated Video Hub & Exam Strategy

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1. Course Units & High-Yield Technical Invariants

Unit / Code	Subject Area & Core Physical / Electronic Topics	Standard Reference & Exam Scoring Triggers
Unit I Solid State	Free electron theory; Fermi energy & Fermi-Dirac distribution $f(E)$; Band theory of solids; Effective mass; Hall effect derivation ($V_H = BI/net$, $R_H = 1/ne$); Direct vs Indirect bandgap; Solar cell Fill Factor & efficiency.	V.K. Mehta Ch 1 / S.O. Pillai - Derivation: $e E_H = e v_d B$ - Solar cell: $FF = (V_{mp} I_{mp}) / (V_{oc} I_{sc})$.
Unit II Electricity & Devices	Ohm's law, KCL, KVL, Voltage & Current division; PN diode characteristics & rectifiers (Half-wave $\eta=40.6\%$, Full-wave Bridge $\eta=81.2\%$); BJT CE mode (α, β); CMOS inverter zero static power; Optical fiber NA = $\sqrt{(n_{core}^2 - n_{clad}^2)}$; CPU vs GPU.	Boylestad Ch 1, 2, 3, 4 - Fiber: $NA = \sin(\theta_a) = \sqrt{(n_{core}^2 - n_{clad}^2)}$ - BJT: $\beta = \alpha / (1 - \alpha)$, $I_E = I_B + I_C$.
Unit III Logic & K-Maps	Number system conversions; 2's complement subtraction & overflow; Binary-to-Gray code ($G_i = B_{(i+1)} \oplus B_i$); Universal NAND/NOR gate synthesis; Boolean algebra; 2/3/4-variable K-Maps with Don't Cares ('X').	Floyd Ch 2, 3, 4 / Mano - K-Map: Octet > Quad > Pair - Universal NAND: XOR requires 4 NANDs.
Unit IV Combinational	Half/Full Adders ($S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$); Half/Full Subtractors; Multiplexers (4:1, 8:1 MUX logic generators); Decoders (3-to-8); Priority Encoders; 2-bit Magnitude Comparators ($A > B$, $A = B$, $A < B$).	Floyd Ch 5, 6 - Full Adder = 2 HAS + 1 OR gate - Comparator: $(A=B) = (A \oplus B)'$ · $(A \oplus B)$'.
Unit V Sequential	Latches vs Flip-Flops (SR, JK, D, T); Master-Slave JK race-around fix; Flip-Flop conversions; Shift Registers (SISO, SIPO, PISO, PIPO); Asynchronous (Ripple) UP/DOWN counters; Mod-N counter with active-LOW clear.	Floyd Ch 7, 8 / R.P. Jain - JK Equation: $Q_{next} = J Q' + K' Q$ - Max Frequency: $f_{max} = 1 / (n \cdot t_{pd})$.
Unit VI Arduino & Sensors	ATmega328P architecture (16 MHz, 14 digital, 6 PWM, 6 analog); 10-bit ADC resolution ($5V/1023 \approx 4.88 \text{ mV}$); Ultrasonic HC-SR04 distance formula ($d = T/58 \text{ cm}$); LDR voltage divider; DHT11 40-bit protocol.	Richard Blum / Tero Karvinen - Ultrasonic: $d = (T \times 0.034 \text{ cm}/\mu s) / 2$ - DHT11: 16-bit H + 16-bit T + 8-bit Checksum.

2. Curated Video Channels & 4-Tier Scoring Strategy

Topic / Unit	Handpicked Video Channel & Key Concepts	100-Percentile Defense Answer
Unit I (Physics)	Dr. Gajendra Purohit / NPTEL: Hall Effect full derivation ($V_H = BI/net$) and Fermi-Dirac distribution.	Draw 3D slab, write Lorentz force balance $e \cdot E_H = e \cdot v_d \cdot B$, and define $R_H = 1/ne$.
Unit II (Devices)	All About Electronics: Diode rectifiers, BJT CE modes, CMOS inverter, Optical Fiber NA.	State Bridge rectifier efficiency (81.2%), PIV (V_m), and derive $NA = \sqrt{(n_{core}^2 - n_{clad}^2)}$.
Unit III (Logic)	Neso Academy: 2's complement arithmetic, Gray codes, 4-variable K-Maps with Don't Cares.	Group octets and quads with wrap-around rules and write minimal SOP expression.
Unit IV (Combinational)	Neso Academy: Full Adder (2 HAS + OR), MUX logic generators, 2-bit Comparator.	Draw gate schematics with truth tables and label XOR, AND, OR components clearly.
Unit V (Sequential)	Neso Academy: Master-Slave JK race-around fix, FF conversion table method.	Construct conversion truth table with required/available excitation columns.
Unit VI (Arduino)	How To Mechatronics: ATmega328P, HC-SR04 distance formula, LDR, DHT11 packet.	State ADC step voltage (4.88 mV) and write ultrasonic formula $d = (T \times 0.034)/2 \text{ cm}$.